Continuous Random Variables

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MATH 241

Week 9, Lecture #1 — Ch. 5.1

Leading to the Exam

1. 10/26-30
   - Continuous random variables
   - To-do: project proposal by 10/30
2. 11/2-6
   - Moments and moment generating functions
   - To-do: HW by 11/2, CP5 on 11/3
3. 11/9-13
   - **Tuesday, 11/10**: Exam #2
   - To-do: HW by 11/9

Feedback from Mid-Semester Survey

- Practice problems and examples have been useful
  - More examples during lecture, fewer proofs (if possible)
  - More worksheets
- More time on problem sets
  - Homework will be due on Monday of the following week

Random Variables - A Recap

$$X \sim f_X(x)$$

- $f_X(x)$ can be constructed by-hand or from a common distribution
  - e.g. Binomial, Geometric
- PMF $f_X(x)$ for point probabilities, CDF $F_X(x)$ for lower-tailed cumulative probabilities
  - $f_X(x) = P(X = x)$; $F_X(x) = P(X \leq x)$
  - Example in R: `dbinom(x, params)`; `pbinom(x, params)`

From Discrete to Continuous

We now care about X as a continuous random variable!

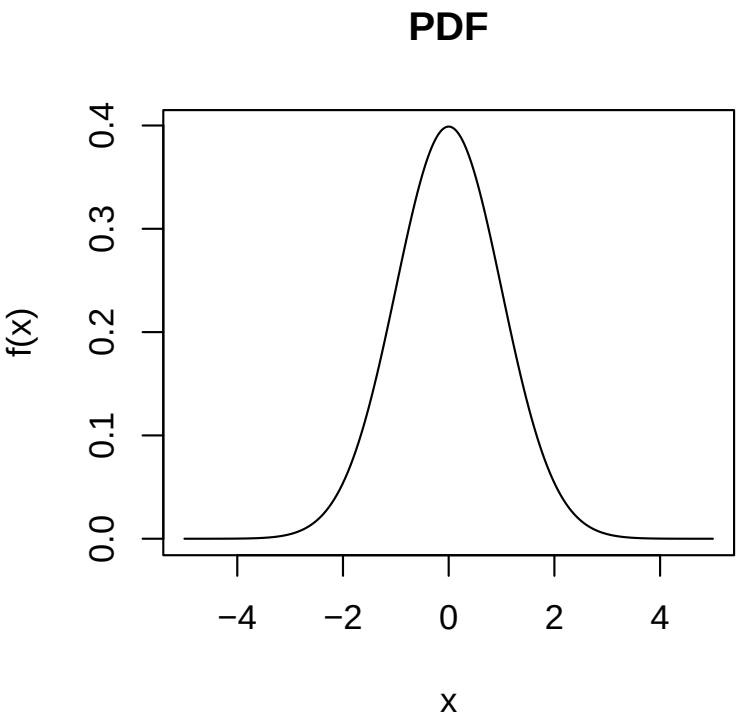
- Probability exists between two bounds a and b
  - Possible values exist in $\mathbb{R}$
  - Consequence of $P(X = x)$?
- Area under the curve
  - Integration
- e.g. time, height, survival

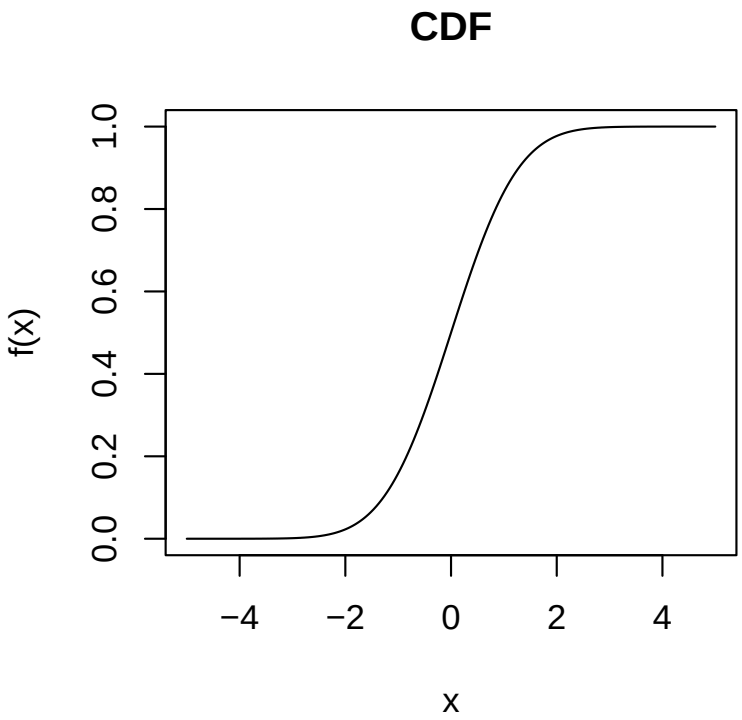
Motivating Examples

Examples of a continuous RV:

- Time in minutes until a train arrives to the platform
- Survival time in months for a cancer patient undergoing treatment
- Cost of medical expenditures in the U.S. in US dollars
- Gene expression levels
- Distance in miles traveled by a car before it breaks down
- Annual income of college graduates in US dollars

Visuals





Continuous RV and a Probability Density Function (PDF)

X is a continuous random variable if there exists a non-negative function f_X defined for all real $x \in (-\infty, \infty)$, such that for all a, $b \in \mathbb{R}$,

$$P(a < X \leq b) = \int_a^b f_X(x)\, dx$$

- f_X is called the **probability density function (pdf)** of X
- Support of X: set of all x where $f_X(x) > 0$

Cumulative Density Function (CDF) and its PDF

The CDF of X is given by:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)dt$$

The PDF of X can be derived from the CDF by taking its derivative:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Recall: What makes a valid PDF? Look through Theorem 5.1.5 for more information.

Questions to Consider

- Is there a difference between these four terms?
  - $P(a \leq X \leq b)$
  - $P(a \leq X < b)$
  - $P(a < X \leq b)$
  - $P(a < X < b)$
- How would you interpret $f_X(x)$ and $F_X(x)$?
- Suppose $f_X(x)$ is piecewise. How does that change how you find $F_X(x)$?
- What changes with $E(X)$ and $Var(X)$?

Example #1

Suppose we care about the time until the next text message comes in from one of your family members, which is denoted as X. Say X has the following PDF:

$$f_X(x) = \frac{3}{2}e^{-\frac{3}{2}x} \qquad \text{for } x \geq 0$$

Objective: Find $F_X(x)$, then use $F_X(x)$ to find these probabilities:

- $P(X < 2)$
- $P(1 < X < 3)$
- $P(X > 3)$

Example #1: CDF

(1) Find $F_X(x)$:

$$\begin{aligned} F_X(x) &= \int_{-\infty}^{x} f_X(t)\,dt \\ &= \int_{-\infty}^{0} 0\,dt + \int_{0}^{x} \frac{3}{2} e^{-\frac{3}{2}t}\,dt \\ &= \int_{0}^{x} \frac{3}{2} e^{-\frac{3}{2}t}\,dt \end{aligned}$$

Example #1: CDF

(1) Find $F_X(x)$:

$$\begin{aligned} F_X(x) &= \int_{-\infty}^{x} f_X(t)\,dt \\ &= \int_{-\infty}^{0} 0\,dt + \int_{0}^{x} \frac{3}{2} e^{-\frac{3}{2}t}\,dt \\ &= \int_{0}^{x} \frac{3}{2} e^{-\frac{3}{2}t}\,dt \end{aligned}$$

Recall: u-substitution

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

- Need to find $u = g(x)$ such that we can properly integrate

Example #1: CDF

(1) Find $F_X(x)$: Let $u = -\frac{3}{2}t$. Then, $du = -\frac{3}{2}dt$ and:

$$
\begin{aligned}
F_X(x) &= \int_0^x \frac{3}{2} e^{-\frac{3}{2}t}\, dt \\
&= -\int_{g(0)}^{g(x)} e^u\, du \\
&= -\left[e^u\right]\Big|_{g(0)}^{g(x)} \\
&= -\left[e^{-\frac{3}{2}t}\right]\Big|_0^x \\
&= -\left[e^{-\frac{3}{2}x} - \cancel{e^{\frac{3}{2}(0)}}\right]^1 \\
&= \begin{cases} 1 - e^{-\frac{3}{2}x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}
\end{aligned}
$$

Example #1: Find Probabilities

Now with our CDF, we can find the following probabilities:

(2) $P(X < 2)$:

$$\begin{aligned} P(X < 2) &= P(X \leq 2) \\ &= F_X(2) \\ &= 1 - e^{-\frac{3}{2}(2)} \\ &= 0.9502 \end{aligned}$$

Example #1: Find Probabilities

Now with our CDF, we can find the following probabilities:

(3) $P(1 < X < 3)$:

$$\begin{aligned} P(1 < X < 3) &= P(1 < X \leq 3) \\ &= F_X(3) - F_X(1) \\ &= 1 - e^{-\frac{3}{2}(3)} - \left[1 - e^{-\frac{3}{2}(1)}\right] \\ &= 0.9889 - 0.7769 \\ &= 0.2120 \end{aligned}$$

Example #1: Find Probabilities

Now with our CDF, we can find the following probabilities:

(4) $P(X > 3)$:

$$\begin{aligned} P(X > 3) &= 1 - P(X \leq 3) \\ &= 1 - F_X(3) \\ &= 1 - \left[1 - e^{-\frac{3}{2}(3)}\right] \\ &= 1 - 0.9889 \\ &= 0.0111 \end{aligned}$$

Activity: Practice

Find the **CDF**, $P(X < 1)$, $P(2 < X < 4)$, and $P(X > 2)$ for X, which has the following (legitimate) PDF:

$$f_X(x) = \begin{cases} \frac{e^x}{e^5 - 1}, & \text{for } 0 \leq x < 5 \\ 0, & \text{otherwise} \end{cases}$$

Bonus: Try to take the derivative of your CDF to get to the PDF.

Things to Consider

- Review certain calculus methods
  - U-substitution for integration
  - Integration by parts
  - L'Hopital's rule
  - Improper integrals
  - Switching order of integration